Manual No 2

AIM

To build a Python-based tool that analyzes the histogram of an input image and automatically adjusts its brightness and contrast to produce a visually balanced image.

OBJECTIVES

- To understand the concept of image histogram and intensity distribution
- To analyze brightness and contrast using histogram properties
- To implement automatic brightness and contrast adjustment
- To enhance image quality using histogram stretching
- To accept an image as user input and display results

PREREQUISITES

- Basic knowledge of **digital image processing**
- Understanding of **grayscale intensity levels (0–255)**
- Familiarity with **Python programming**
- Basic knowledge of **NumPy and OpenCV**
- Concept of **histogram, brightness, and contrast**

THEORY

Image Histogram

An image histogram represents the frequency of pixel intensity values in an image.

- X-axis → Intensity values (0 to 255)
- Y-axis → Number of pixels

Histogram analysis helps in understanding:

- **Brightness** → overall lightness/darkness of image
- **Contrast** → spread of intensity values

Brightness

Brightness depends on the **average intensity** of an image.

- Dark image → histogram shifted left

- Bright image → histogram shifted right

Contrast

Contrast depends on the **range of intensity values**.

- Low contrast → histogram clustered in narrow range
- High contrast → histogram spread across full range

Automatic Brightness & Contrast Adjustment

Automatic adjustment can be done using **histogram stretching (normalization)**:

$$I_{new} = \frac{(I - I_{min})}{(I_{max} - I_{min})} \times 255$$

Where:

- I_{min}= minimum intensity in image
- I_{max}= maximum intensity in image

This improves contrast and balances brightness automatically.

PSEUDOCODE

Automatic Brightness and Contrast Adjustment using Histogram Analysis

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Algorithm: Auto Brightness & Contrast Adjustment

START

1. Input image path from user

2. Read the image from given path

3. IF image is not found

    Display error message

    STOP

  END IF

4. Convert image to grayscale (if it is a color image)

5. Compute histogram of grayscale image

   - Intensity range: 0 to 255

6. Find minimum intensity value (Imin) in the image

7. Find maximum intensity value (Imax) in the image

8. FOR each pixel intensity I in the image DO

   Normalize pixel value using:

   I_new = (I - Imin) / (Imax - Imin) × 255

  END FOR

9. Convert normalized pixel values to integer range (0–255)

10. Generate histogram of adjusted image

11. Display:

   a) Original image

   b) Brightness and contrast adjusted image

12. Plot:

   a) Histogram of original image

   b) Histogram of adjusted image

END